

Classical Mechanics

$$W = \vec{F} \cdot d\vec{r}$$

$$(\vec{F}_c) d\vec{r}$$

Force of constraint is missing.

$$(\vec{F}_e + \vec{F}_c) \cdot d\vec{r}$$

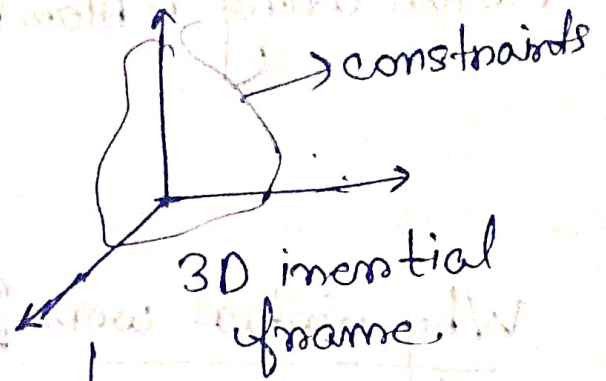
$$\frac{d^2 \vec{r}}{dt^2} = (\vec{F}_e + \vec{F}_c)$$

Virtual work

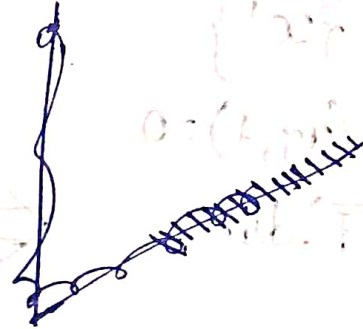
defined as

virtual displacement from one pt. to another
(displacement of a system keeping everything frozen (time) also)

transformation } generalized coordinates
constraints } (constraints will not appear here).



configuration space,



Assignment

Derive the ~~equations of~~ Lagrange of equation of Motion using D'Alembert's Principle.

→ Submit - Friday.

20/08/2025

Why virtual work?

Inertial Frame

$$\begin{aligned} &\vec{r}_i \\ &\{x^i\} \\ &f(\vec{r}_i, t) = 0 \\ &\vec{F} \cdot d\vec{r} \end{aligned}$$

Generalized Coordinates

$$\begin{aligned} &\{q_i\} \\ &\text{no constraint} \end{aligned}$$

$$\begin{aligned} &Q_i dq_i \\ &\hookrightarrow Q_i \text{ is generalized force.} \end{aligned}$$

$$\vec{F} = \sum_i (F_i \hat{x}_i) = \sum_i Q_i \hat{q}_i$$

$$\begin{aligned} x^i &= x^i(q_j, t) \\ dx^i &= \sum_j \frac{\partial x^i}{\partial q_j} dq_j + \frac{\partial x^i}{\partial t} dt \end{aligned}$$

$$dW = \sum_{i,j} F_i \frac{\partial x^i}{\partial q_j} dq_j + \sum_i F_i \frac{\partial x^i}{\partial t} dt = \sum_j Q_j dq_j +$$

$$\delta x^i = \frac{\partial x^i}{\partial q_j} \delta q_j + \left(\dots \right)$$

→ Time is not frozen → instantaneously calculate in δt .

• A particle of mass 'm' is confined to a circle which is expanding with time and subjected to a force $\vec{F} = F_x \hat{x} + F_y \hat{y}$.

$$x^2 + y^2 = R^2(t).$$

$$\Rightarrow x = R(t) \cos \theta.$$

$$y = R(t) \sin \theta.$$

$$dx = -R(t) \sin \theta d\theta + \frac{\partial R}{\partial t} \cos \theta dt.$$

$$dy = R(t) \cos \theta d\theta + \frac{\partial R}{\partial t} \sin \theta dt.$$

$$dW = F_i dx^i = F_x \left[-R(t) \sin \theta d\theta + \frac{\partial R}{\partial t} \cos \theta dt \right] + F_y \left[R(t) \cos \theta d\theta + \frac{\partial R}{\partial t} \sin \theta dt \right].$$

$$dW = \underbrace{[-F_x \sin \theta + F_y \cos \theta]}_{\text{form of } Q} R d\theta + \underbrace{[F_x \cos \theta + F_y \sin \theta]}_{\Pi} \frac{\partial R}{\partial t} dt$$

Work should be in the form of $Q d\theta$. But here additional Π term, which we cannot equate to 0 as $dR \neq 0$ and F_x and $F_y \neq 0$ as they are function of x and y and $\theta \neq 0$ $\sin \theta, \cos \theta \neq 0$ always. → This is a contradiction.

Lagrange Equation of Motion

① We have generalized coordinate $(q_1, q_2, \dots, q_n)$.

② Kinetic energy $T(q_i, \dot{q}_i, t) \rightarrow \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$.

③ Potential $V(q_1, q_2, \dots, t)$, $F_j = -\frac{\partial V}{\partial q_j}$.

Define

$$L = T - V.$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} = 0.$$

$\rightarrow$ 2nd order Coupled Equation.

Cartesian Coordinates $\{x, y, z\}$ $T = \frac{1}{2} m (\dot{x})^2$

$$\frac{d}{dt} (m\dot{x}) + \frac{\partial V}{\partial x} = 0$$

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x}$$

$$\Rightarrow m\ddot{x} + \frac{\partial V}{\partial x} = 0$$

Plane Polar Coordinate

$$x = r \cos \theta, \quad y = r \sin \theta.$$

$$T = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) = \frac{m}{2} [\dot{r}^2 + r^2 \dot{\theta}^2]$$

$$m \frac{dr}{dt} - m r \dot{\theta}^2 \quad \boxed{\frac{\partial L}{\partial r} = m r \dot{\theta}^2}$$

$$[F_r = m\ddot{r} - m r \dot{\theta}^2]$$

$$\frac{\partial L}{\partial r} = m \dot{\theta}^2$$

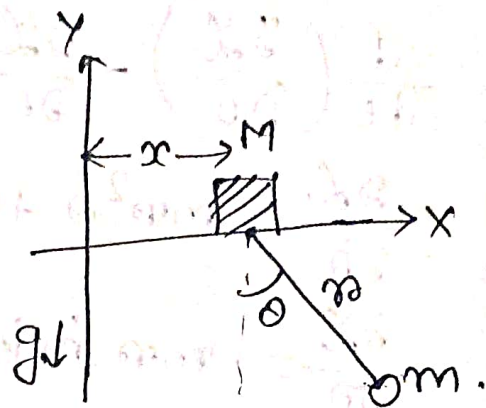
$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m r^2 \dot{\theta} \quad \frac{\partial \mathcal{L}}{\partial \theta} = 0$$

$$\frac{d}{dt} (m r^2 \dot{\theta}) = 2 m r \dot{r} \dot{\theta} + m r^2 \ddot{\theta} = F_{\theta}.$$

$$2 m r \dot{r} \dot{\theta} + m r^2 \ddot{\theta} = F_{\theta}.$$

Gen

Generalized coordinates
are $\{x, \theta\}$.



kinetic energy of M :- $\frac{1}{2} M \dot{x}^2$
(T_M)

For the small mass m ,
displacement $\rightarrow$

$$x = r \sin \theta, \quad \dot{x} = r \cos \theta \dot{\theta}$$

$$y = r \cos \theta, \quad \dot{y} = -r \sin \theta \dot{\theta}$$

~~$\frac{1}{2} m \dot{x}^2 + \frac{1}{2} m \dot{y}^2$~~
 $x = r \sin \theta$
 $y = r \cos \theta$

$$T_m = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2)$$

$$\dot{x}' = \dot{x} + r \cos \theta \dot{\theta}$$

$$\dot{y}' = -r \sin \theta \dot{\theta}$$

$$\Rightarrow T_m = \frac{1}{2} m (\dot{x}^2 + 2r \cos \theta \dot{x} \dot{\theta} + r^2 \cos^2 \theta \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\theta}^2)$$

$$= \frac{1}{2} m (\dot{x}^2 + r^2 \dot{\theta}^2 + 2r \cos \theta \dot{x} \dot{\theta})$$

Potential energy of m (V_m) $= -mg r \cos \theta$

$$\mathcal{L} = T - V \quad T = T_m + T_M$$

$$\Rightarrow \mathcal{L} = \frac{1}{2} (M+m) \dot{x}^2 + \frac{1}{2} m (r^2 \dot{\theta}^2 + 2r \dot{x} \cos \theta \dot{\theta})$$

$$+ mg r \cos \theta$$

$$\mathcal{L}(x, \dot{x}, \theta, \dot{\theta})$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) - \frac{\partial \mathcal{L}}{\partial x} = 0$$

$$\frac{\partial \mathcal{L}}{\partial x} = (M+m) \dot{x} + m r \cos \theta \dot{\theta}, \quad \frac{\partial \mathcal{L}}{\partial x} = 0$$

$$\Rightarrow \frac{d}{dt} \left[(M+m) \dot{x} + mn \cos \theta \dot{\theta} \right] = 0$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) - \frac{\partial \mathcal{L}}{\partial \theta} = 0$$

$$\hookrightarrow (M+m) \dot{x} + mn \cos \theta \dot{\theta} = \text{constant} \quad (\text{Ans})$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = mn^2 \dot{\theta} + mn \dot{x} \cos \theta$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = -mn \dot{x} \sin \theta - mgn \sin \theta$$

$$\Rightarrow \frac{d}{dt} (mn^2 \dot{\theta} + mn \dot{x} \cos \theta) + mn \dot{x} \sin \theta \dot{\theta} + mgn \sin \theta = 0$$

$$\Rightarrow mn^2 \ddot{\theta} + mn \ddot{x} \cos \theta - mn \dot{x} \sin \theta \dot{\theta} + mn \dot{x} \sin \theta \dot{\theta} + mgn \sin \theta = 0$$

$$\Rightarrow mn^2 \ddot{\theta} + mn \ddot{x} \cos \theta + mgn \sin \theta = 0$$

$$\Rightarrow n \ddot{\theta} + \ddot{x} \cos \theta + g \sin \theta = 0 \quad (\text{Ans})$$

$$2) \mathcal{L} = T - V$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = 0 \quad \mathcal{L}' = \mathcal{L} + \frac{dF(q,t)}{dt}$$

$$\mathcal{L}' = \mathcal{L} + \frac{\partial F}{\partial q_j} \dot{q}_j + \frac{\partial F}{\partial t}$$

$$\frac{\partial \mathcal{L}'}{\partial \dot{q}_j} = \frac{\partial \mathcal{L}}{\partial \dot{q}_j} + \frac{\partial F}{\partial q_j}$$

$$\frac{\partial \mathcal{L}'}{\partial q_j} = \frac{\partial \mathcal{L}}{\partial q_j} + \frac{\partial^2 F}{\partial q_k \partial q_j} \dot{q}_j + \frac{\partial^2 F}{\partial q_j \partial t} \Rightarrow \frac{d}{dt} \left(\frac{\partial \mathcal{L}'}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}'}{\partial q_j} = \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j}$$

Gauge Degree of Lagrangian:-

LEOM is invariant under transformation $\mathcal{L}' = \mathcal{L} + \frac{dF(q,t)}{dt}$

where $F(q,t)$ is a function of (q,t) only.

Ex:- $\mathcal{L} = \frac{1}{2} q \dot{q}$, $F = -\frac{1}{4} q^2$

$$\Rightarrow \frac{dF}{dt} = -\frac{1}{2} q \dot{q} \Rightarrow \mathcal{L}' = 0 \rightarrow \text{Not a Lagrangian}$$

$$\mathcal{L} = T - V$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) = -Q_j$$

$$-\frac{\partial V}{\partial q_j} = Q_j$$

$$\frac{dP_j}{dt} = Q_j$$

Generalized momentum $P_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i}$

Ex:-

$$\mathcal{L} = \frac{1}{2} m \dot{x}^2$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = m \dot{x} = P_x$$

$$\mathcal{L} = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2)$$

$$P_r = m \dot{r}, P_\theta = m r^2 \dot{\theta}$$

In a Lagrangian system -

$$\mathcal{L}(q, \dot{q}, t) \Rightarrow p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i}$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = 0.$$

$$\text{if } q_j, \frac{\partial \mathcal{L}}{\partial q_j} = 0.$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) = 0$$

$$p_j = \frac{\partial \mathcal{L}}{\partial \dot{q}_j} = \text{constant}.$$

cyclic coordinate.

- If Lagrangian is independent of a specific generalized coordinate $q_i \Rightarrow \frac{\partial \mathcal{L}}{\partial q_i} = 0$, corresponding generalized p_i is constant of motion.

$$\frac{dp_j}{dt} = 0.$$

q_j is called cyclic coordinate.

$$\mathcal{L} = \frac{1}{2} m \dot{x}^2 \quad \frac{\partial \mathcal{L}}{\partial \dot{x}} = p_x = \text{constant}.$$

$$\mathcal{L} = \frac{1}{2} (m \dot{r}^2 + r^2 \dot{\theta}^2)$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = 0 \quad p_\theta = m r^2 \dot{\theta}$$

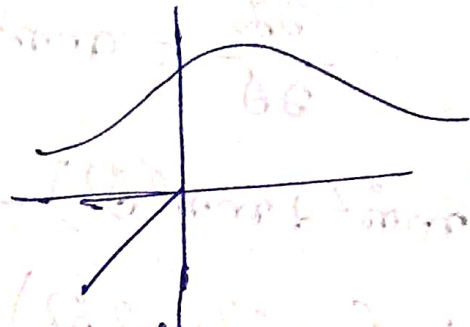
can these be

$$\frac{d\mathcal{L}}{dt} = 0 \text{ ?}$$

$$\mathcal{L}(q, \dot{q}, t)$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = 0$$

$$\mathcal{L}(q, \dot{q}, t) = \text{const.}$$



$$\frac{d\mathcal{L}}{dt} = \sum_i \frac{\partial \mathcal{L}}{\partial q_i} \dot{q}_i + \sum_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \frac{d}{dt}(\dot{q}_i) + \frac{\partial \mathcal{L}}{\partial t}$$

$$= \sum_j \left[\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) \dot{q}_j + \frac{\partial \mathcal{L}}{\partial \dot{q}_j} \frac{d}{dt}(\dot{q}_j) \right] + \frac{\partial \mathcal{L}}{\partial t}$$

$$\frac{d\mathcal{L}}{dt} = \frac{d}{dt} \left(\sum \dot{q}_j \frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) + \frac{\partial \mathcal{L}}{\partial t}$$

$$\frac{d}{dt} \left[\left(\sum \dot{q}_j \frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \mathcal{L} \right] = - \frac{\partial \mathcal{L}}{\partial t}$$

$$h = \sum \dot{q}_j \frac{\partial \mathcal{L}}{\partial \dot{q}_j} - \mathcal{L}$$

$$h = \sum \dot{q}_j p_j - \mathcal{L}$$

h is called energy function

$$\frac{dh}{dt} = - \frac{\partial \mathcal{L}}{\partial t}$$

If Lagrangian $\mathcal{L}$ is not an explicit function of time, then energy function h is constant of motion.

$$h = \sum \dot{q}_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \mathcal{L}$$

$$L = \frac{1}{2} [m\dot{r}^2 + mr^2\dot{\theta}^2] - V(r).$$

$$\frac{\partial L}{\partial t} = 0, \quad \frac{\partial L}{\partial \dot{r}} = m\dot{r}.$$

$$\frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta}$$

$$h = (m\dot{r}^2 + mr^2\dot{\theta}^2) - \frac{1}{2} m (\dot{r}^2 + mr^2\dot{\theta}^2) + V.$$

$$h = \frac{1}{2} (m\dot{r}^2 + mr^2\dot{\theta}^2) + V.$$

$$= T + V = E \rightarrow \text{Total energy}$$

$$L = T(q, \dot{q}, t) - V(q).$$

In Euclidean coordinates -

$$T = \sum_i \frac{1}{2} m_i v_i^2.$$

$$\vec{v}_i = \frac{d\vec{r}_i}{dt}$$

$$\vec{v}_i = \sum_k \frac{\partial \vec{r}_i}{\partial q_k} \dot{q}_k + \frac{\partial \vec{r}_i}{\partial t}$$

$$\begin{aligned} \sum_i \frac{1}{2} m_i v_i^2 &= \sum_i \frac{1}{2} m_i \left(\sum_j \frac{\partial \vec{r}_i}{\partial q_j} \dot{q}_j + \frac{\partial \vec{r}_i}{\partial t} \right)^2 \\ &= \sum_i \frac{1}{2} m_i \sum_{j,k} \left(\frac{\partial \vec{r}_i}{\partial q_j} \dot{q}_j \right) \left(\frac{\partial \vec{r}_i}{\partial q_k} \dot{q}_k \right) \\ &\quad + m_i \frac{\partial \vec{r}_i}{\partial t} \sum_j \frac{\partial \vec{r}_i}{\partial q_j} \dot{q}_j + \frac{1}{2} m_i \frac{\partial \vec{r}_i}{\partial t} \cdot \frac{\partial \vec{r}_i}{\partial t} \end{aligned}$$

$$= \frac{1}{2} \sum_{j,k} M_{jk} \dot{q}_k \dot{q}_j + \sum_j M_j \dot{q}_j + M_0$$

$$M_{ij} = \sum_i m_i \frac{\partial \vec{r}_i}{\partial q_j} \cdot \frac{\partial \vec{r}_i}{\partial q_k}.$$

$$M_j = \sum_i m_i \frac{\partial m_i}{\partial q_j} \frac{\partial m_i}{\partial t}$$

$$M_0 = \frac{1}{2} \sum m_i \left(\frac{\partial m_i}{\partial t} \right)^2$$

$$T = T_0 + T_1 + T_2$$

$T_0 \Rightarrow$ function of q, t only.

$T_1 \Rightarrow$ linear homogeneous function in $\dot{q}$.

Coefficient can be function of q and t .

$T_2 \Rightarrow$ 2nd order homogeneous function in $\dot{q}_i$.

$$L = T - V(q_i)$$

$$L(q, \dot{q}, t) = L_0(q, t) + L_1(q, \dot{q}, t) + L_2(q, \dot{q}, t).$$

$$L = L_0 + L_1 + L_2.$$

L_0 : is function of q, t .

L_1 : is function of q, t and linear homogeneous in $\dot{q}_i$.

L_2 : is ^{also} function of q, t and second order homogeneous in $\dot{q}_i$.

If $f(x_i)$ is homogeneous function of order n .

Then we have -

$$\sum_i x_i \frac{\partial f}{\partial x_i} = n f$$

$$h = \sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - L.$$

from here.

$$L = L_0 + L_1 + L_2$$

$$h = \sum_i \dot{q}_i \frac{\partial}{\partial \dot{q}_i} (L_0 + L_1 + L_2) - (L_0 + L_1 + L_2).$$

$$h = L_1 + 2L_2 - L_0 - L_1 - L_2.$$

$$h = L_2 - L_0.$$

When h becomes ^{total} energy? his a function of energy (energy function).
_{condition}

$$\Rightarrow \text{When } \boxed{L_2 = T, L_0 = -V.}$$

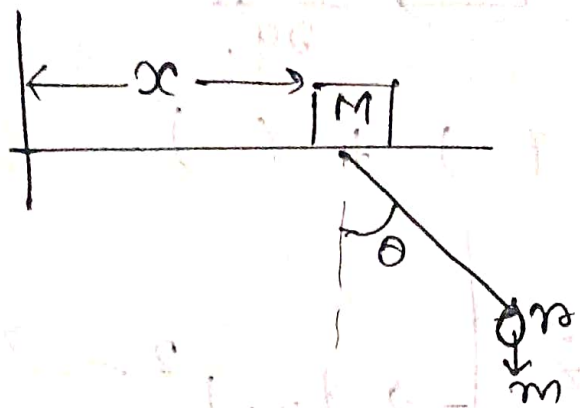
$$\text{and } T = \vec{p}_0^0 + \vec{p}_1^0 + T_2 \Rightarrow T = T_2.$$

$$L = \frac{1}{2} \sum () \dot{q}_i \dot{q}_j \quad L_0(q, t) = -V.$$

$$T = \frac{1}{2} \sum_i m_i v_i^2$$

Example :-

1)



Lagrangian (done previously) —

$$\mathcal{L}(x, \dot{x}, \theta, \dot{\theta}) = \frac{1}{2} (M+m) \dot{x}^2 + \frac{m}{2} (r^2 \dot{\theta}^2 + 2r \dot{\theta} \dot{x} \cos \theta) + mgr \cos \theta$$

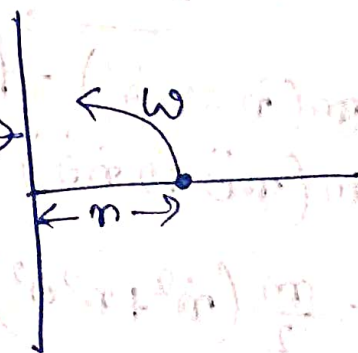
$$T = \frac{1}{2} (M+m) \dot{x}^2 + \frac{m}{2} (r^2 \dot{\theta}^2 + 2r \dot{\theta} \dot{x} \cos \theta)$$

$$T = T_2 - \mathcal{L}_2$$

$$\mathcal{L}_0 = -V = mgr \cos \theta$$

$$h = T + V$$

2) Example from Goldstein.



$$T = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m r^2 \omega^2 = T_0 + T_2$$

$$V = 0$$

ω is constant.

$$h = \mathcal{L}_2 - \mathcal{L}_0 = \frac{1}{2} m \dot{r}^2 - \frac{1}{2} m r^2 \omega^2$$

$$h = \sum_i \dot{q}_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \mathcal{L}$$

$$E = T + V = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m r^2 \omega^2$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) - \frac{\partial \mathcal{L}}{\partial q} = 0$$

$$\rightarrow \frac{\partial V_{eff}}{\partial q}$$

$$h = \sum \dot{q}_i \dot{q}_j + () + ()$$

$$L_x = y \dot{z} - z \dot{y}$$

$$L_y = z \dot{x} - x \dot{z}$$

$$L_z = x \dot{y} - y \dot{x}$$

$$L_z, L^2$$

they commute

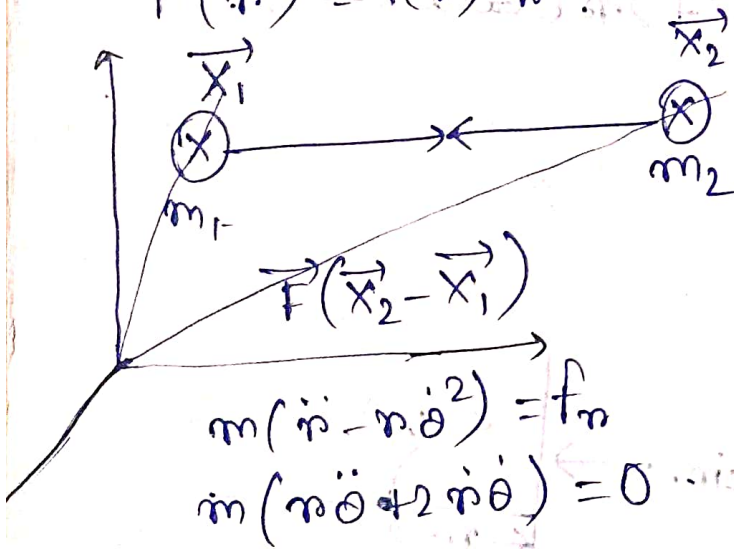
conserved and function of quadratic w.r.t. in velocity.

27/08/2025

Central Force Problem

$$\vec{F}(\vec{r}) = f(r) \hat{r}$$

$$\vec{F}(\vec{r}) = -\nabla V = f(r) \hat{r} = V(r)$$



$$\mathcal{L} = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2) - V(r)$$

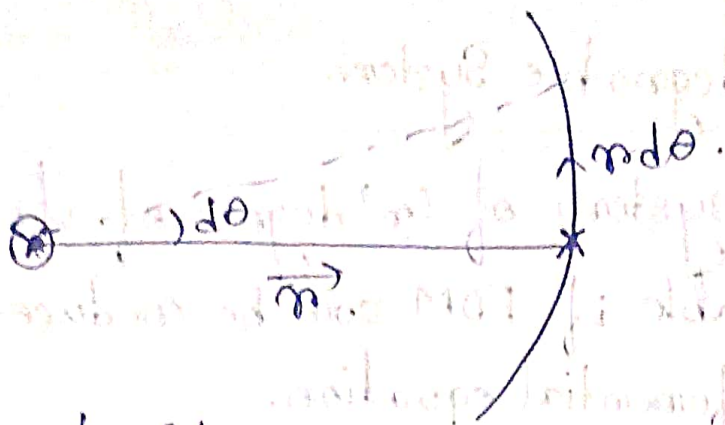
$$\frac{\partial \mathcal{L}}{\partial \theta} = 0 \quad \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = L_z = m r^2 \dot{\theta} = \text{constant}$$

$$\dot{\theta} = \frac{L_z}{m r^2}$$

$$\frac{d}{dt} (m r^2 \dot{\theta}) = 0$$

$$\frac{d}{dt} m \frac{d}{dt} \left(\frac{1}{2} r^2 \dot{\theta} \right) = 0$$

$$\text{Angular velocity} = \frac{d\theta}{dt}$$



$$dA = \frac{1}{2} r^2 d\theta$$

$$\Rightarrow \frac{dA}{dt} = \frac{1}{2} r^2 \dot{\theta}$$

$$\Rightarrow \frac{dA}{dt} = \text{constant}$$

$$V(r) = -\frac{GMm}{r} \Rightarrow \text{Kepler's Problem}$$

$$\frac{\partial \mathcal{L}}{\partial t} = 0, \quad h = L_2 - L_0$$

$$L_2 = T, \quad L_0 = -V$$

$$h = T + V = E$$

$$E = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2) + V(r)$$

$$\frac{2}{m} (E - V) = \dot{r}^2 + r^2 \dot{\theta}^2$$

$$= \dot{r}^2 + \frac{L_z^2}{m^2 r^2}$$

$$\dot{r}^2 = \frac{2}{m} (E - V) - \frac{L_z^2}{m^2 r^2}$$

Integrable System

If a Lagrangian system of (n') degree of freedom is said to be integrable if EOM can be reduced to (n') first order differential equation.

Corollary:- If Lagrangian system has (n) independent constant of motion then it is integrable.

$$\dot{r} = \pm \left[\frac{2}{m} (E - V) - \frac{L_z^2}{m^2 r^2} \right]^{\frac{1}{2}}$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{r}} \right) - \frac{\partial \mathcal{L}}{\partial r} = 0.$$

$$\frac{\partial \mathcal{L}}{\partial \dot{r}} = m \dot{r} \quad \frac{\partial \mathcal{L}}{\partial r} = m r \dot{\theta}^2 - \frac{\partial V}{\partial r}$$

$$m \ddot{r} - m r \dot{\theta}^2 + \frac{\partial V}{\partial r} = 0.$$

$$\Rightarrow m \ddot{r} - \frac{L_z^2}{m r^3} + \frac{\partial V}{\partial r} = 0$$

$$\Rightarrow m \ddot{r} = \frac{L_z^2}{m r^3} - \frac{\partial V}{\partial r} = - \frac{\partial}{\partial r} \left(\frac{L_z^2}{2 m r^2} + V \right).$$

$$m \ddot{r} = F_r = - \nabla V_{\text{eff}}.$$

$$V_{\text{eff}} = \frac{L_z^2}{2 m r^2} + V.$$

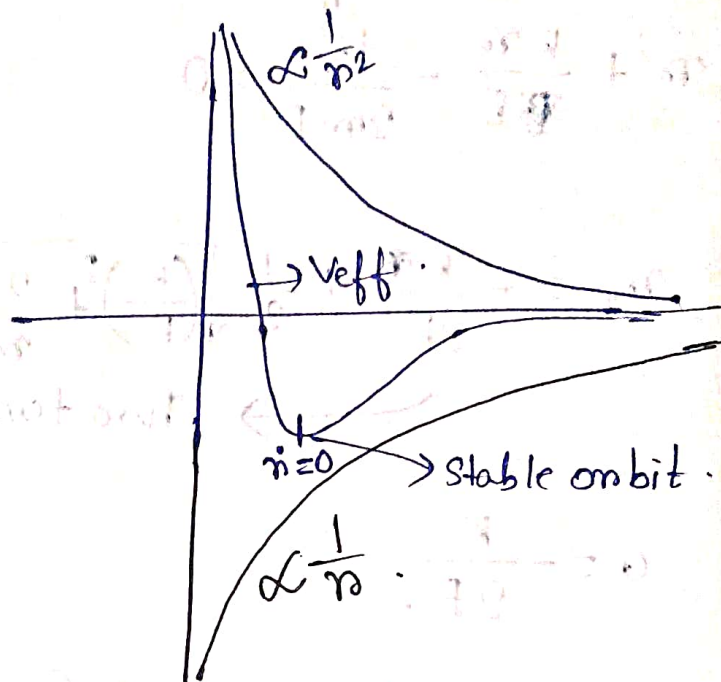
$$\dot{r} = \pm \left[\frac{2}{m} \left(E - V - \frac{L_z^2}{2 m r^2} \right) \right]^{\frac{1}{2}}$$
$$= \pm \left[\frac{2}{m} (E - V_{\text{eff}}) \right]^{\frac{1}{2}}$$

Kepler Problem

$$V_{\text{eff}} = \frac{L_z^2}{2mr^2} - \frac{k}{r}$$

$$V_{\text{kep}} = -\frac{k}{r}$$

$$k = G M m$$



29/08/2025

Central Force Problem

$$\mathcal{L} = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2) - V(r)$$

$$m r^2 \dot{\theta} = L_z \Rightarrow \dot{\theta} = \frac{L_z}{m r^2}$$

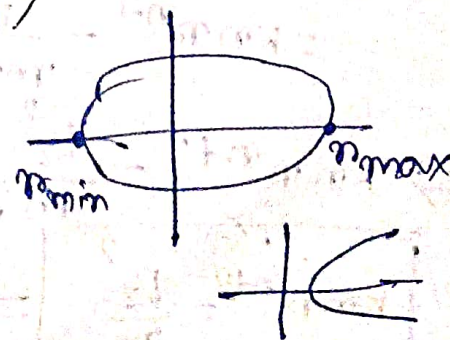
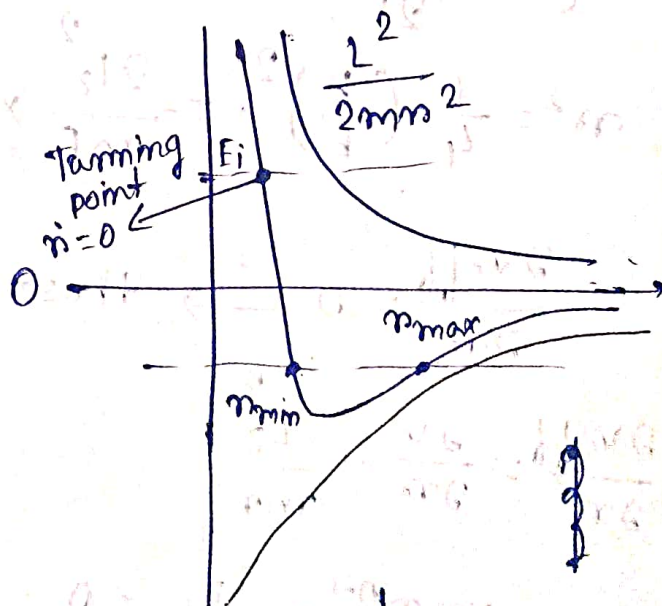
$$\dot{r} = \left[\frac{2}{m} \left(E - V - \frac{L_z^2}{2m r^2} \right) \right]^{\frac{1}{2}}$$

$$V_{\text{eff}} = V + \frac{1}{2} \frac{L_z^2}{m r^2}$$

$$E = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2) + V$$

Kepler problem: $V(r) = -\frac{k}{r}$

Semi major axis $= (r_{\text{min}} + r_{\text{max}}) \times \frac{1}{2} = a$



$$1 + \frac{k}{r} - \frac{L_z^2}{2mr^2} = 0$$

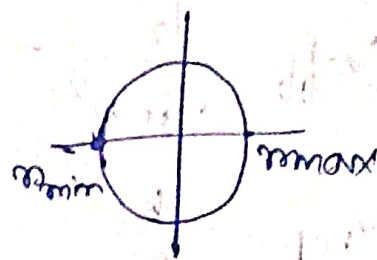
$$r^2 + \frac{k r}{E} - \frac{L_z^2}{2mE} = 0$$

$$r_{\pm} = -\frac{k}{2E} \pm \frac{1}{2} \sqrt{\left(\frac{k}{E}\right)^2 + \frac{2L_z^2}{mE}}$$

Two turning point.

$$a = -\frac{k}{2E}$$

Circular Orbit

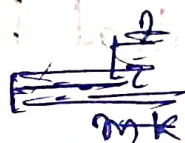


$$r_{\max} = r_{\min}$$

For circular orbit (r_c):-

$$r_+ = r_- \Rightarrow \frac{k^2}{E^2} = -\frac{2L_z^2}{mE}$$

We have taken $E < 0$.



$$r_c = -\frac{k}{2E}$$

$$r^2 = \frac{1}{4} \left(\frac{k}{E}\right)^2 = -\frac{2L_z^2}{mE} \times \frac{1}{4} = -\frac{1}{2} \frac{L_z^2}{mE}$$

check whole derivation once.

$\frac{\partial V_{eff}}{\partial r} = 0 \rightarrow$ these are the optimum points.

$$\begin{aligned} \frac{\partial V_{eff}}{\partial r} &= \frac{\partial V}{\partial r} - \frac{L_z^2}{mr^3} \\ &= kmr^{n-1} - \frac{L_z^2}{mr^3} = 0 \end{aligned}$$

$$\Rightarrow kmr^{n+2} = \frac{L_z^2}{m}$$

when $k = -\frac{GMm}{r}$ we go back to the Kepler problem.

$$\boxed{kmr^{n+2} = \frac{L_z^2}{m}} \Rightarrow \text{condition for circular orbit.}$$

$$V \propto km^n$$

Any function can be expressed as the power of r^n .

For stable circular orbit

$$\frac{\partial^2 V}{\partial r^2} > 0$$

$$\Rightarrow km(n-1)r^{n-2} + \frac{3L_z^2}{mr^4} > 0$$

$$\Rightarrow km(n-1)r^{n+2} + \frac{3L_z^2}{m} > 0$$

$$\frac{L_z^2}{m} = kmr^{n+2}$$

$$\Rightarrow km(n-1)r^{n+2} + 3kmr^{n+2} > 0$$

$$\Rightarrow (n-1+3)\frac{L_z^2}{m} > 0$$

$$\Rightarrow (n+2)\frac{L_z^2}{m} > 0$$

Orbit equation $r(\theta)$

$\theta(t), r(t)$

$$\frac{dr}{dt} / \frac{d\theta}{dt} = \frac{dr}{d\theta}$$

$$\frac{d\theta}{dt} = \frac{L_z}{mr^2} \Rightarrow dt = mr^2 \frac{d\theta}{L_z}$$

$$m\ddot{r} = -\frac{\partial}{\partial r} \left[V + \frac{1}{2} \frac{L_z^2}{mr^2} \right] = -\frac{d}{dr} \left[V + \frac{1}{2} \frac{L_z^2}{mr^2} \right]$$

$$\frac{d}{dt} \left(m \frac{dr}{dt} \right) \frac{d\theta}{dt} = -\frac{d}{dr} \left[V + \frac{1}{2} \frac{L_z^2}{mr^2} \right] \frac{d\theta}{dt} \frac{dt}{d\theta}$$

$$\Rightarrow \frac{d}{d\theta} \left(\frac{1}{r^2} \frac{dr}{d\theta} \right) = \frac{d}{dr} \left[V + \frac{1}{2} \frac{L_z^2}{mr^2} \right] \frac{m}{L_z}$$

$$U = \frac{1}{r}, \quad \frac{dU}{d\theta} = -\frac{1}{r^2} \frac{dr}{d\theta} = \frac{dU}{dr} \frac{dr}{d\theta}$$

By calculating explicitly we will get -

$$\frac{d}{d\theta} \left(\frac{1}{r} \frac{dU}{d\theta} \right) = \frac{mk}{L_z^2} - U$$

$$\frac{v^2}{u^2} = \frac{1}{n^2}$$

Homework :-

Show that

$$m\ddot{r} = \frac{d}{dr} \left[-\frac{k}{r} + \frac{L_z^2}{2mr^2} \right]$$

Change $\frac{dr}{dt}$ to $\frac{dr}{d\theta}$.
and $v \Rightarrow \frac{1}{n}$

Finally we have,

$$\frac{d}{d\theta} \left(\frac{du}{d\theta} \right) + u = \frac{mk}{L_z^2}$$

$$\Rightarrow \frac{d^2 u}{d\theta^2} + u = \frac{mk}{L_z^2}$$

$$\Rightarrow u = A \cos(\theta - \theta_0) + \frac{mk}{L_z^2} \quad \left(\text{Harmonic Oscillation type equation} \right)$$

$$\frac{1}{r} = \frac{mk}{L_z^2} \left[1 + \sqrt{\frac{1 + 2EL_z^2}{mk^2}} \cos(\theta - \theta_0) \right]$$

$$\frac{1}{r} = C [1 + e \cos(\theta - \theta_0)] \rightarrow \text{equation of ellipse}$$

$$e = \sqrt{\frac{1 + 2EL_z^2}{mk^2}} = \text{eccentricity of ellipse}$$

Homework :- (Next Friday).

Show that bound orbit in keplers problem are ellipses. Find the expression for e in terms of l_z, E .

- GAP -

Hamiltonian Formalism (Legendre's Transformation).

$$\mathcal{L}(q, \dot{q}, t)$$

$$q \longrightarrow q'$$

$$\dot{q} \longrightarrow \dot{q}'$$

In Lagrangian we don't have freedom to write generalized velocity in ~~the~~ the transformation equation with generalized coordinates. Like,

$$x^2 + y^2 = r^2$$

but we can not write $\dot{x}^2 + \dot{y}^2 = \dots$

although generalized velocities are independent.

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}} \right) - \frac{\partial \mathcal{L}}{\partial q} = 0$$

$$\therefore \frac{d}{dt} (p) = \frac{\partial \mathcal{L}}{\partial q}$$

Replace $\dot{q}$ by $p = \frac{\partial \mathcal{L}}{\partial \dot{q}}$.

$f(x)$ replace x by $u = \frac{\partial f}{\partial x}$.

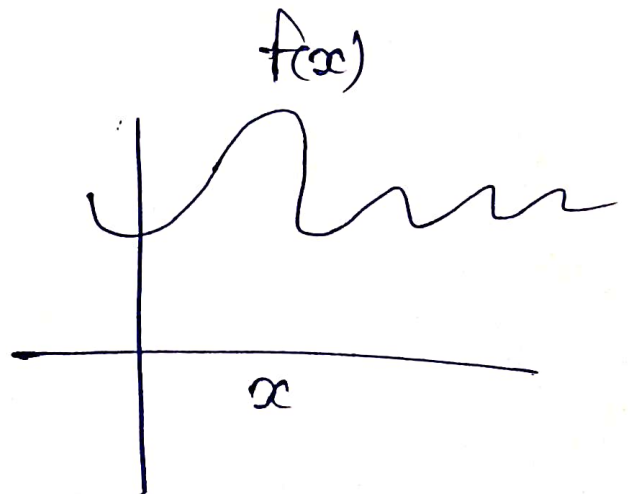
$$df = \frac{\partial f}{\partial x} dx = u dx$$

$$g = f - ux$$

$$dg = df - (du)x - u dx$$

$$\text{assume } g \equiv g(u) \Rightarrow dg = \frac{\partial g}{\partial u} du \Rightarrow dg = -x du$$

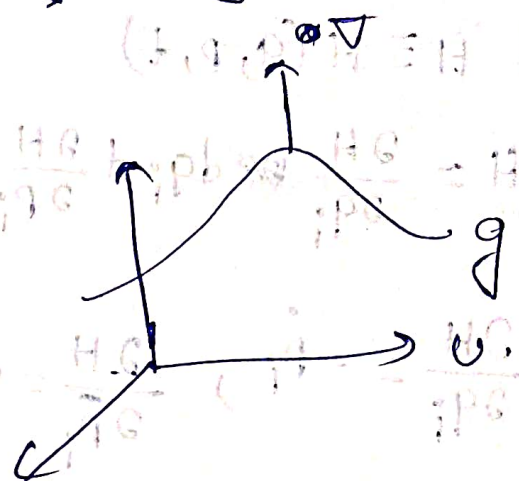
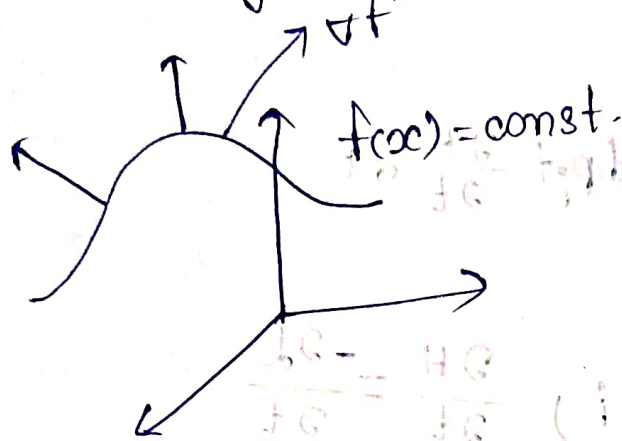
$$\frac{\partial g}{\partial u} = -x$$



$$\left\{ \begin{matrix} f(x) \\ x \end{matrix} \right\} \rightarrow \left\{ \begin{matrix} g \\ u \end{matrix} \right\}$$

$$g = f - ux$$

$\frac{\partial g}{\partial u} = -x$, then $g(u)$ function of u only.



$$df = \nabla f \cdot dx$$

$$\left\{ \begin{matrix} f(x) \\ x \end{matrix} \right\} \rightarrow \left\{ \begin{matrix} g(u) \\ u \end{matrix} \right\}$$

$$x = \dot{q}, u = p = \frac{\partial \mathcal{L}}{\partial \dot{q}}$$

$$\mathcal{L}(q_i, \dot{q}_i, t)$$

$$\dot{q} \rightarrow \frac{\partial \mathcal{L}}{\partial \dot{q}} = p$$

$$H = \sum_j \dot{q}_j p_j - \mathcal{L} = \dot{q}_j \cdot p_j - \mathcal{L}$$

$$d\mathcal{L} = \left(\frac{\partial \mathcal{L}}{\partial q_i} \right) dq_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i} d\dot{q}_i + \frac{\partial \mathcal{L}}{\partial t} dt$$

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) = \frac{\partial \mathcal{L}}{\partial q_i}$$

$$\frac{d}{dt} p_i \equiv \dot{p}_i \Rightarrow d\mathcal{L} = \dot{p}_i dq_i + p_i d\dot{q}_i + \frac{\partial \mathcal{L}}{\partial t} dt$$

$$dH = d\dot{q}_j P_j + \dot{q}_j dP_j - dL.$$

$$dH = d\dot{q}_j P_j + \dot{q}_j dP_j - \dot{P}_j dq_j - P_j d\dot{q}_j - \frac{\partial L}{\partial t} dt$$

$$= \dot{q}_j dP_j - \dot{P}_j dq_j - \frac{\partial L}{\partial t} dt$$

$$\text{Let } H \equiv H(q, p, t)$$

$$dH = \frac{\partial H}{\partial q_i} dq_i + \frac{\partial H}{\partial p_i} dp_i + \frac{\partial H}{\partial t} dt$$

$$\frac{\partial H}{\partial q_i} = -\dot{p}_i, \quad \frac{\partial H}{\partial p_i} = \dot{q}_i, \quad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}$$

$$L(q, \dot{q}, t) \quad \dot{q} \rightarrow \frac{\partial L}{\partial \dot{q}} = p$$

$$H = P_i \dot{q}_i - L, \quad \frac{\partial H}{\partial \dot{q}} = 0$$

$$\dot{p}_i = \frac{dp_i}{dt} = -\frac{\partial H}{\partial q_i}, \quad \dot{q}_i = \frac{\partial H}{\partial p_i}$$

Example :-

$$\textcircled{1} L = \frac{m}{2} (\dot{x}^2 + \dot{y}^2) - V(x).$$

$$H = P_i \dot{q}_i - L.$$

$$\Rightarrow H = \frac{m}{2} (\dot{x}^2 + \dot{y}^2) + V(x).$$

Not a hamiltonian as it explicitly depends on $\dot{x}, \dot{y}$. It is simply an energy function.

$$P_i = \frac{\partial L}{\partial \dot{q}_i}$$

write $\dot{q}_i$ in terms of P_i

$$P_x = m\dot{x} \Rightarrow \dot{x} = \frac{P_x}{m}$$

$$H = \frac{1}{2m} [P_x^2 + P_y^2] + V(x)$$

This is hamiltonian

EOM -

$$\dot{p}_x = -\frac{\partial H}{\partial x} = -\frac{\partial V}{\partial x}.$$

$$\dot{x} = \frac{\partial H}{\partial p_x} = \frac{p_x}{m}.$$

$$\Rightarrow m\ddot{x} = -\frac{\partial V}{\partial x}.$$

$$\dot{p}_i = -\frac{\partial H}{\partial q_i} \quad \text{if} \quad \frac{\partial H}{\partial q_i} = 0 \Rightarrow \frac{dp_i}{dt} = 0 \Rightarrow p_i = \text{constant}.$$

$$\frac{\partial \mathcal{L}}{\partial t} = -\frac{\partial H}{\partial t} = 0$$

$\Rightarrow H$ is constant of motion.

Now,

$$\mathcal{L} = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2) - V(r).$$

$$H = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2) + V(r).$$

$$p_r = m\dot{r}, \quad p_\theta = mr^2\dot{\theta}$$

$$\Rightarrow \dot{r} = \frac{p_r}{m}, \quad \dot{\theta} = \frac{p_\theta}{mr^2}.$$

$$\Rightarrow H = \frac{1}{2m} \left(p_r^2 + \frac{p_\theta^2}{r^2} \right) + V.$$

$$\frac{\partial H}{\partial \theta} = 0 \Rightarrow \dot{p}_\theta = mr^2\dot{\theta} = L_z = \text{constant}.$$

Form of Hamiltonian

Lagrangian takes the form

$$L = L_0 + L_1(q, \dot{q}, t) + L_2(q, \dot{q}, t)$$

$L_1 \rightarrow$ linear homogeneous in $\dot{q}$

$L_2 \rightarrow$ second order homogeneous in $\dot{q}$

$$L = L_0 + a_i \dot{q}_i + \frac{1}{2} T_{ij} \dot{q}_i \dot{q}_j$$

$$T_{ij} = T_{ji}$$

$$H = p_i \dot{q}_i - L$$

$$= p_i \dot{q}_i - L_0 - a_i \dot{q}_i - \frac{1}{2} T_{ij} \dot{q}_i \dot{q}_j$$

$$p_i = \frac{\partial L}{\partial \dot{q}_i} = \frac{\partial}{\partial \dot{q}_i} \left[a_i \dot{q}_i + \frac{1}{2} T_{lm} \dot{q}_l \dot{q}_m \right]$$

$$= a_i + \frac{1}{2} T_{lm} (\delta_{il} \dot{q}_m + \dot{q}_l \delta_{im})$$

$\frac{\partial x_i}{\partial x_j} = \delta_{ij}$

$$p_i = a_i + \frac{1}{2} (T_{lm} \dot{q}_m + T_{li} \dot{q}_l)$$

$$= a_i + \frac{1}{2} (T_{lm} \dot{q}_m + T_{mi} \dot{q}_m) \quad (\text{As dummy variable } l, m)$$

$$\Rightarrow p_i = a_i + T_{im} \dot{q}_m \quad [\because T \text{ symmetric matrix }]$$

$$\Rightarrow p_i - a_i = T_{im} \dot{q}_m$$

$\Rightarrow$

$$A \times B = C$$

$$C_{ik} = A_{ij} B_{jk} = B_{jk} A_{ij}$$

$$A_{ij} \neq A_{ji}, B_{jk} \neq B_{kj}$$

$$A \times B \neq B \times A$$

$$T^{-1} T = 1$$

$$T_{ij}^{-1} T_{jk} = \delta_{ik}$$

$$\Rightarrow (T_{P_i})^{-1}$$

$$\Rightarrow T_{P_i}^{-1} (P_i - a_i) = T_{P_i}^{-1} T_{im} \dot{q}_m = \delta_{pm} \dot{q}_m = \dot{q}_p$$

$$\Rightarrow \dot{q}_p = T_{P_i}^{-1} (P_i - a_i)$$

$$H = (P_i - a_i) \dot{q}_i - \frac{1}{2} T_{ij} \dot{q}_i \dot{q}_j = L_0$$

$$\Rightarrow H = T_{ij}^{-1} (P_i - a_i) (P_j - a_j) - \frac{1}{2} T_{ij}^{-1} T_{il} (P_l - a_l) T_{jm}^{-1} (P_m - a_m) - L_0$$

$$T_{ij} T_{jm}^{-1} = \delta_{im}$$

$$\Rightarrow H = T_{ij}^{-1} (P_i - a_i) (P_j - a_j) - \frac{1}{2} T_{ij}^{-1} (P_j - a_j) (P_i - a_i) - L_0$$

$$H = \frac{1}{2} T_{ij}^{-1} (P_i - a_i) (P_j - a_j) - L_0$$

$$T_{ij}^{-1} \text{ does not exist } \Rightarrow \text{rank}(T_{ij}) \neq \dim(T_{ij})$$

$\Rightarrow$ ^{At least} one generalized coordinate can be written as the form of other, as T_{ij} is kinetic energy.

That cannot happen $\Rightarrow T_{ij}^{-1}$ must exist in this case.

If we start with Lagrangian of the form

$$L = L_0 + a_i \dot{q}_i + \frac{1}{2} T_{ij} \dot{q}_i \dot{q}_j$$

$$H = \frac{1}{2} T_{ij}^{-1} (p_i - a_i) (p_j - a_j) - L_0$$

a_i , T_{ij} , and L_0 are functions of q and t .

$$L = \underbrace{\frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2)}_{\frac{1}{2} T_{ij} \dot{q}_i \dot{q}_j} + \underbrace{m \Omega r^2 \sin^2 \theta \dot{\phi}}_{a_i \dot{q}_i} - \underbrace{V(r)}_{L_0}$$

for a particle in rotating frame with uniform angular velocity at $\hat{z}$ axis with Ω constant

$$\vec{F} = \frac{d\vec{p}}{dt}$$

$$L = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2) - V(r)$$

$$\phi' = \phi + \Omega t, \quad \dot{\phi}' = \dot{\phi} + \Omega$$

$$L = \frac{m}{2} (\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta (\dot{\phi} + \Omega)^2) - V(r)$$

$$T_{ij} = m \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2 \theta \end{pmatrix}$$

$$T_{ij}^{-1} = \frac{1}{m} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & \frac{1}{r^2 \sin^2 \theta} \end{pmatrix}$$

$$a_i = \begin{pmatrix} 0 \\ 0 \\ m \Omega r^2 \sin^2 \theta \end{pmatrix}$$

$$L_0 = \frac{1}{2} m \Omega^2 r^2 \sin^2 \theta - V(r)$$

$$H = \frac{1}{2} T_{ij}^{-1} (p_i - a_i) (p_j - a_j) - h_0$$

$$P = \begin{pmatrix} p_n \\ p_\theta \\ p_\phi \end{pmatrix}$$

$$H = \frac{1}{2} (P-a)^T (T_{ij})^{-1} (P-a) - h_0$$

$$H = \frac{1}{2m} p_n^2 + \frac{1}{2mn^2} p_\theta^2 + \frac{1}{2m n^2 \sin^2 \theta} (p_\phi - \frac{1}{2} m \Omega n^2 \sin^2 \theta)^2 - h_0$$

Simplectic geometry



$$\mathcal{L}(q, \dot{q}, t) \longrightarrow H(q, p, t) \Rightarrow 2m$$

$$\{q\} \longrightarrow \ddot{q}_i \quad \{q, p\}, \quad \begin{aligned} \dot{q}_i &= \frac{\partial H}{\partial p_i} \\ \dot{p}_i &= -\frac{\partial H}{\partial q_i} \end{aligned}$$

$2m \rightarrow$ coordinates, q_i, p_i

$$n_i = q_i, \quad m_{i+m} = p_i, \quad H(n)$$

$$n = \begin{pmatrix} q_1 \\ q_2 \\ \vdots \\ q_m \\ p_1 \\ p_2 \\ \vdots \\ p_m \end{pmatrix}$$

Gradient

$$\frac{\partial H}{\partial n_i} = \frac{\partial H}{\partial q_i}, \quad \frac{\partial H}{\partial n_{i+m}} = \frac{\partial H}{\partial p_i}, \quad i = 1, \dots, m$$

$$\dot{n}_i = \frac{\partial H}{\partial n_i}$$

$$J = \begin{pmatrix} \Phi & \mathbb{1} \\ -\mathbb{1} & \Phi \end{pmatrix} \rightarrow 2n \times 2n \text{ matrix}$$

Here, $\Phi \Rightarrow n \times n$ 0 matrix
 $\mathbb{1} \Rightarrow$ unit matrix in n -D.

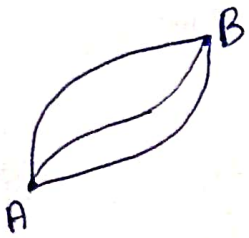
$$\tilde{J} = J^{-1} = -J^T \quad (|J| = +1) \quad \frac{1}{2} \dot{q}^T \dot{q} = H$$

$$\dot{q}_i = J_{ij} \frac{\partial H}{\partial p_j}$$

i, j from 1 to $2n$.

26/09/2025

Hamiltonian Principle and Hamiltonian



$$\frac{\partial L}{\partial \dot{q}} \rightarrow (p, q, p) \rightarrow (p, q, p)$$

Principle of least action -

$$\delta I = \int L(q, \dot{q}, t) dt \Rightarrow \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0.$$

$$H = p_i \dot{q}_i - L \Rightarrow L = p_i \dot{q}_i - H(q, p, t).$$

$$L(q, \dot{q}_i, p, \dot{p}_i, t) = p_i \dot{q}_i - H = f.$$

$$\Rightarrow \delta I = 0 = \delta \int f(q, \dot{q}, p, \dot{p}, t) dt.$$

$$\frac{d}{dt} \left(\frac{\partial f}{\partial \dot{q}_i} \right) - \frac{\partial f}{\partial q_i} = 0$$

$$\frac{d}{dt} \left(\frac{\partial f}{\partial \dot{p}_i} \right) - \frac{\partial f}{\partial p_i} = 0.$$

$$\frac{\partial f}{\partial \dot{q}_i} = p_i, \quad \frac{\partial f}{\partial q_i} = -\frac{\partial H}{\partial q_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial p_i}$$

$$\delta \int L(q, \dot{q}, t) = 0$$

$$\Rightarrow \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = 0$$

$$\Rightarrow \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{p}_i} \right) - \frac{\partial \mathcal{L}}{\partial p_i} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \dot{q}_i} = 0 \Rightarrow \frac{\partial \mathcal{L}}{\partial p_i} = 0$$

$$\frac{\partial \mathcal{L}}{\partial p_i} = \dot{q}_i - \frac{\partial H}{\partial p_i} \Rightarrow \dot{q}_i = \frac{\partial H}{\partial p_i}$$

$$\{q, p\} \rightarrow \{Q, P\}$$

$$\left(\begin{matrix} \{q, p\} \\ H \end{matrix} \right) \rightarrow \text{Canonical variable also } H$$

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

Canonical variable $\{q, p, H\}$ such that transformation from one set of canonical variables to another set of canonical variable is called canonical transformation.

$$\{q, p, H\} \iff \{Q, P, K\}$$

$$\dot{q}_i = \frac{\partial H}{\partial p_i} \quad \text{Canonical Transformation}$$

$$\dot{p}_i = -\frac{\partial H}{\partial q_i}$$

$$\dot{Q} = \frac{\partial K}{\partial P_i}$$

$$\dot{P}_i = -\frac{\partial K}{\partial Q_i}$$

$$q_i = q_i(Q_i, P_i, t)$$

$$p_i = p_i(Q_i, P_i, t)$$

$$S\Gamma = 0 = S \int (p_i \dot{q}_i - H) dt = 0$$

$$\Downarrow$$

$$S\Gamma = S \int (p_i \dot{Q}_i - K) dt = 0$$

$$\lambda (p_i \dot{q}_i - H) = p_i \dot{Q}_i - K + \frac{dF(p, q, P, Q, t)}{dt}$$

generating function.

Let choose $\lambda = 1$.

Consider a transformation,

$$q_i = \alpha Q_i$$

$$p_i = \beta P_i, \quad \alpha \beta = \lambda.$$

λ acts on H , which is not significant.

So, when $\lambda \neq 1$ we always can do an transformation such that $\lambda = 1$.

canonical variable $\{q_i, p_i, H\}$.

if they satisfy HEOM $\rightarrow \dot{q}_i = \frac{\partial H}{\partial p_i}, \dot{p}_i = -\frac{\partial H}{\partial q_i}$

$$\mathcal{I} = \int_A^B (p_i \dot{q}_i - H) dt \text{ such that } \delta \mathcal{I} = 0 \Rightarrow \text{HEOM}$$

Canonical Transformation :-

Transformation from one set of canonical variable to another set is called canonical Transformation.

$$\{q_i, p_i, H\} \rightarrow \{Q_i, P_i, K\}$$

$$\dot{q}_i = \frac{\partial H}{\partial p_i} \text{ and } \dot{Q}_i = \frac{\partial K}{\partial P_i}$$

$$\dot{p}_i = -\frac{\partial H}{\partial q_i} \text{ and } \dot{P}_i = -\frac{\partial K}{\partial Q_i}$$

$$\delta \mathcal{I} = \delta \int [P_i \dot{Q}_i - K] dt = \delta \int [P_i \dot{Q}_i - K] dt$$

$$P_i \dot{Q}_i - H = P_i \dot{Q}_i - K + \frac{dF}{dt} (q, p, Q, P, t)$$

$$F = F_1(q_i, Q_i, t)$$

$$P_i \dot{Q}_i - H = P_i \dot{Q}_i - K + \frac{dF_1}{dt}$$

$$\Rightarrow P_i \dot{Q}_i - H = P_i \dot{Q}_i - K + \frac{\partial F_1}{\partial t} + \frac{\partial F_1}{\partial q_i} \dot{q}_i + \frac{\partial F_1}{\partial Q_i} \dot{Q}_i$$

$$P_i = \frac{\partial F}{\partial q_i}, \quad P_i = -\frac{\partial F}{\partial Q_i}$$

$$K = H + \frac{\partial F_1}{\partial t}$$

$$F = F_2(q_i, p_i, t) \rightarrow F = F_2(q_i, p_i, t) - P_i Q_i$$

$$P_i \dot{Q}_i - H = P_i \dot{Q}_i - K + \frac{dF_2}{dt} \Rightarrow \cancel{P_i \dot{Q}_i} - K$$

$$\Rightarrow P_i \dot{Q}_i - H = P_i \dot{Q}_i - K + \frac{\partial F_2}{\partial t} + \frac{\partial F_2}{\partial q_i} \dot{q}_i + \frac{\partial F_2}{\partial p_i} \dot{p}_i$$

$$P_i \dot{q}_i - H = P_i \dot{Q}_i - K + \frac{\partial F_2}{\partial t} + \frac{\partial F_2}{\partial q_i} \dot{q}_i + \frac{\partial F_2}{\partial p_i} \dot{p}_i - P_i \dot{Q}_i - \hat{P}_i \dot{Q}_i$$

$$P_i = \frac{\partial F_2}{\partial \dot{q}_i}, \quad Q_i = \frac{\partial F_2}{\partial \dot{p}_i}$$

$$P_i \dot{Q}_i - K = 0 \quad \text{and} \quad \frac{\partial F_2}{\partial t} + \frac{\partial F_2}{\partial q_i} \dot{q}_i + \frac{\partial F_2}{\partial p_i} \dot{p}_i - P_i \dot{Q}_i = 0$$

Canonical transformation

Transformation from one set of canonical variables to another set is called canonical transformation.

$$(x, p, t) \rightarrow (X, P, T)$$

$$\frac{\partial X}{\partial p} = \dot{x} \quad \text{and} \quad \frac{\partial X}{\partial p} = \dot{p}$$

$$\frac{\partial X}{\partial p} = \dot{x} \quad \text{and} \quad \frac{\partial X}{\partial p} = \dot{p}$$

$$\frac{\partial}{\partial t} [X - \dot{x} p] = 0 \quad \text{and} \quad \frac{\partial}{\partial t} [X - \dot{x} p] = 0$$

$$(X, P, T) = F(x, p, t) + \text{const}$$

$$F(x, p, t) = F(x, p, t)$$

$$\frac{\partial F}{\partial p} + \dot{x} - \dot{p} = 0 \quad \text{and} \quad \frac{\partial F}{\partial p} + \dot{x} - \dot{p} = 0$$

$$\dot{p} \frac{\partial F}{\partial p} + \dot{p} \frac{\partial F}{\partial p} + \frac{\partial F}{\partial t} + \dot{x} - \dot{p} = 0 \quad \text{and} \quad \dot{p} \frac{\partial F}{\partial p} + \dot{p} \frac{\partial F}{\partial p} + \frac{\partial F}{\partial t} + \dot{x} - \dot{p} = 0$$

$$\frac{\partial F}{\partial p} = \dot{x} \quad \text{and} \quad \frac{\partial F}{\partial p} = \dot{p}$$

$$\frac{\partial F}{\partial p} + \dot{x} - \dot{p} = 0 \quad \text{and} \quad \frac{\partial F}{\partial p} + \dot{x} - \dot{p} = 0$$

$$F(x, p, t) = F(x, p, t) \quad \text{and} \quad F(x, p, t) = F(x, p, t)$$

$$\frac{\partial F}{\partial p} + \dot{x} - \dot{p} = 0 \quad \text{and} \quad \frac{\partial F}{\partial p} + \dot{x} - \dot{p} = 0$$

$$\dot{p} \frac{\partial F}{\partial p} + \dot{p} \frac{\partial F}{\partial p} + \frac{\partial F}{\partial t} + \dot{x} - \dot{p} = 0 \quad \text{and} \quad \dot{p} \frac{\partial F}{\partial p} + \dot{p} \frac{\partial F}{\partial p} + \frac{\partial F}{\partial t} + \dot{x} - \dot{p} = 0$$

Set of canonical variables $\{q, p, H\}$

another set of canonical variables $\{Q, P, K\}$

relation,

$$-P_i \dot{q}_i + H = -P_i \dot{Q}_i + K + \frac{dF}{dt}$$

generating function :-

$$q, p \rightarrow Q, P$$

$$F = F_1(q, Q, t)$$

$$P_i = \frac{\partial F_1}{\partial q_i} \text{ and } P_i = -\frac{\partial F_1}{\partial Q_i}$$

$$F = F_2(q, P, t) - Q_i P_i$$

$$P_i = \frac{\partial F_2}{\partial q_i} \text{ and } Q_i = \frac{\partial F_2}{\partial P_i}$$

$$F = F_3(p, Q, t) + q_i P_i$$

$$q_i = -\frac{\partial F_3}{\partial P_i} \text{ and } P_i = -\frac{\partial F_3}{\partial Q_i}$$

$$F = F_4(p, P, t) + q_i P_i - Q_i P_i$$

$$q_i = -\frac{\partial F_4}{\partial P_i} \text{ and } Q_i = \frac{\partial F_4}{\partial P_i}$$

2-D system :-

$$q_1, q_2, p_1, p_2$$

$$F'(q_1, p_2, p_1, Q_2, t)$$

$$F = F' + q_2 p_2 - p_1 Q_1$$

$$P_1 = \frac{\partial F}{\partial q_1}, Q_1 = \frac{\partial F'}{\partial p_1}, q_2 = -\frac{\partial F'}{\partial p_2}, P_2 = \frac{\partial F'}{\partial Q_2}$$

$$F_2 = f_j(q_1, \dots, q_m, t) P_j$$

$$Q_i = \frac{\partial F_2}{\partial P_j} = \delta_{ij} f_j(q_1, \dots, q_m, t) = f_i(q_1, \dots, q_m, t)$$

$$P_i = \frac{\partial f_i}{\partial q_j} P_j$$

let's make so change F_2 .

$$F_2 = f_j(q_1, \dots, q_m, t) P_j + g(q_1, \dots, q_m, t)$$

Q_i will be same.

$$\text{But, } P_i = \frac{\partial f_j}{\partial q_j} P_j + \frac{\partial g}{\partial q_i}$$

$$H = \frac{P^2}{2m} + \frac{kq^2}{2}$$

$$H = \frac{1}{2m} [P^2 + m^2 \omega^2 q^2] = K$$

$$\omega^2 = \frac{k}{m}$$

$\{q, P, H\} \rightarrow \{Q, P, K\}$ make transformation

such that Q is cyclic.

We have to make two variable to one variable.

Transform p and q in mind

$$P = f(P) \sin Q \cos Q$$

$$q = (f(P) \cos Q) / m\omega$$

$$H = \frac{1}{2m} f^2(p) [\cos^2 Q + \sin^2 Q] = \frac{f^2(p)}{2m} = K.$$

$$p = f(p) \cos Q, \quad q = \frac{f(p) \sin Q}{m\omega}$$

$$F_1(q, Q), \quad p = \frac{\partial F_1}{\partial q}, \quad p = -\frac{\partial F_1}{\partial Q}$$

$$\frac{p}{q} = \frac{m\omega \cos Q}{\sin Q}$$

$$p = m\omega q \frac{\cos Q}{\sin Q} = \frac{\partial F_1}{\partial q}$$

$$F_1 = \frac{m\omega}{2} q^2 \frac{\cos Q}{\sin Q}$$

$$p = -\frac{\partial F_1}{\partial Q} = \frac{m\omega q^2}{2 \sin^2 Q}$$

$$p = f(p) \cos Q = m\omega q \frac{\cos Q}{\sin Q}$$

$$q^2 = \frac{2p}{m\omega} \sin^2 Q$$

$$q = \sqrt{\frac{2p}{m\omega}} \sin Q$$

$$f(p) = \sqrt{2m\omega p}$$

$$H = \omega p$$

p is constant of motion, $p = \frac{E}{\omega}$ (let).

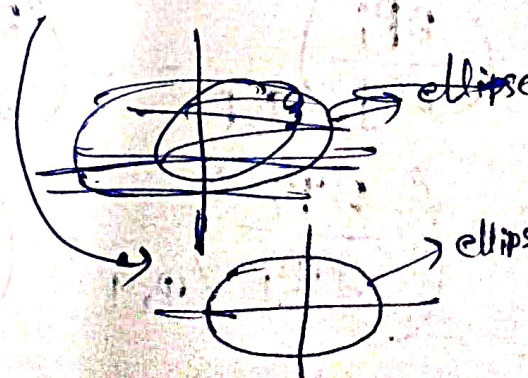
$$\Rightarrow H = E, \quad Q = \omega t + \alpha$$

$$\dot{Q} = \frac{\partial H}{\partial p} = \omega$$

$$q = \sqrt{\frac{2E}{m\omega^2}} \sin(\omega t + \alpha)$$

$$p = \sqrt{2mE} \cos(\omega t + \alpha)$$

p vs q graph



Symplectic Geometry and Canonical Transformation

$$\eta = \begin{pmatrix} q_i \\ p_i \end{pmatrix}, \quad i=1, \dots, m$$

$$\eta_i = \begin{pmatrix} q \\ p \end{pmatrix}, \quad i=1, \dots, 2m$$

$$\dot{\eta}_i = J_{ij} \frac{\partial H}{\partial \eta_j}$$

$$J = \begin{bmatrix} 0 & \mathbb{1} \\ -\mathbb{1} & 0 \end{bmatrix}$$

$$\tilde{J} = J^{-1} = -J = \begin{bmatrix} 0 & -\mathbb{1} \\ \mathbb{1} & 0 \end{bmatrix}$$

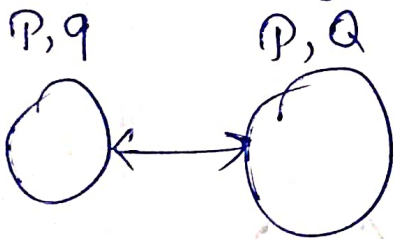
$0 \rightarrow$ ~~identity~~ zero matrix in dimension
 $\mathbb{1} \rightarrow$ identity

$$Q_i = Q_i(q, p), \quad P_i = P_i(q, p)$$

$$\Leftrightarrow q_j = q_j(Q, P), \quad p_j = p_j(Q, P)$$

$$\dot{Q}_i = \frac{\partial Q_i}{\partial q_j} \dot{q}_j + \frac{\partial Q_i}{\partial p_j} \dot{p}_j = \frac{\partial Q_i}{\partial q_j} \frac{\partial H}{\partial p_j} - \frac{\partial Q_i}{\partial p_j} \frac{\partial H}{\partial q_j}$$

$$\dot{Q}_i = \frac{\partial H}{\partial P_j} = \frac{\partial H}{\partial P_j} \frac{\partial p_j}{\partial P_i} + \frac{\partial H}{\partial q_j} \frac{\partial q_j}{\partial P_i}$$



$$\left(\frac{\partial Q_i}{\partial q_j} \right)_{q, p} = \left(\frac{\partial P_j}{\partial P_i} \right)_{Q, P}$$

$$\left(\frac{\partial Q_i}{\partial p_j} \right)_{q, p} = - \left(\frac{\partial q_j}{\partial P_i} \right)_{Q, P}$$

$$\left(\frac{\partial P_i}{\partial q_j} \right)_{q, p} = - \left(\frac{\partial p_j}{\partial Q_i} \right)_{Q, P}$$

$$\left(\frac{\partial P_i}{\partial p_j} \right)_{q, p} = \left(\frac{\partial q_j}{\partial Q_i} \right)_{Q, P}$$

$P \rightarrow$ capital p

$$\{q, p\} \rightarrow \{Q, P\}.$$

$$\begin{cases} q = q(Q, P) \\ p = p(Q, P) \end{cases}$$

$$\begin{cases} Q = Q(q, p) \\ P = P(q, p) \end{cases}$$

Symplectic Geometry

$$x_{ii} = \begin{pmatrix} q \\ p \end{pmatrix}$$

$$\dot{x}_i = J_{ij} \frac{\partial H}{\partial x_j}$$

$$J = J^{-1} = -J.$$

$$\dot{q}_i = \frac{\partial \mathcal{L}_i}{\partial p_j} \dot{p}_j = M_{ij} \dot{p}_j \quad M_{ij} = \frac{\partial \mathcal{L}_i}{\partial p_j}$$

$$\dot{q}_i = M_{ij} J_{jk} \frac{\partial H}{\partial p_k}$$

$$\dot{q}_i = J_{ij} \frac{\partial H}{\partial p_j} = J_{ij} \frac{\partial H}{\partial p_k} \frac{\partial p_k}{\partial \mathcal{L}_j}$$

$$M_{ij} J_{jk} \frac{\partial H}{\partial p_k} = J_{ij} M_{jk}^{-1} \frac{\partial H}{\partial p_k}$$

$$J = M J \tilde{M}$$

$$\{q, p\} \Rightarrow x_i \Rightarrow \mathcal{L}_i$$

$$\{q, p\} \Rightarrow \mathcal{L}_i$$

$$\tilde{M} \tilde{M}^T = \mathbb{I}$$

$$\dot{q}, \dot{p} \in \tilde{M} = \mathbb{I}$$

$$\tilde{M} = \begin{pmatrix} \frac{\partial q}{\partial Q} & \frac{\partial p}{\partial Q} \\ \frac{\partial q}{\partial P} & \frac{\partial p}{\partial P} \end{pmatrix}$$

$$\tilde{M} \tilde{M}^T = \mathbb{I}$$

$$x \longleftrightarrow \mathcal{L}_i, \quad \mathcal{L}_i = \mathcal{L}_i(x)$$

$$\tilde{M}^{-1} = \tilde{M}^T, \quad (\tilde{M}^T = M^T)$$

$$\tilde{M}^{-1} = \tilde{M}^T, \quad (\tilde{M}^T = M^T)$$

$$\tilde{M} = \begin{pmatrix} \frac{\partial q}{\partial Q} & \frac{\partial p}{\partial Q} \\ \frac{\partial q}{\partial P} & \frac{\partial p}{\partial P} \end{pmatrix}$$

$$\tilde{M} = \begin{pmatrix} \frac{\partial q}{\partial Q} & \frac{\partial p}{\partial Q} \\ \frac{\partial q}{\partial P} & \frac{\partial p}{\partial P} \end{pmatrix}$$

$$\tilde{M} \tilde{M}^T = \mathbb{I}, \quad \tilde{M}^T \tilde{M} = \mathbb{I}$$

$$n_i = n_i(\xi).$$

$$J = M J \tilde{M}$$

$$M_{ij} = \frac{\partial \xi_i}{\partial n_j}$$

$$J = \tilde{M} J M$$

Poisson Bracket

Any two functions u, v of canonical variables

(q, p) is defined as

$$[u, v]_{p, q} = \left[\frac{\partial u}{\partial q_i} \frac{\partial v}{\partial p_i} - \frac{\partial v}{\partial q_i} \frac{\partial u}{\partial p_i} \right]$$

$$[u, v]_n = \frac{\partial u}{\partial n_i} J_{ij} \frac{\partial v}{\partial n_j} = \frac{\partial u}{\partial n} J \frac{\partial v}{\partial n}$$

$$[q_j, q_k]_{q, p} = 0 = [p_j, p_k]_{q, p}$$

$$[q_j, p_k]_{p, q} = \frac{\partial q_j}{\partial q_i} \frac{\partial p_k}{\partial p_i} - \frac{\partial p_k}{\partial q_i} \frac{\partial q_j}{\partial p_i}$$

$$= -[p_k, q_j]_{p, q} = \delta_{ji} \delta_{ki} = \delta_{jk}$$

$$[n_i, n_j]_n = J_{ij}$$

$$[n, n]_n = J$$

Given $\{q, p\} = n$ Find $\{Q, P\}$ canonical or not?

$$n_i = n_i(\xi) \quad \text{and} \quad \xi_i = \xi_i(n)$$

$$M_{ij} = \frac{\partial \xi_i}{\partial n_j} = M$$

$$\tilde{M} J M = J, \quad M J \tilde{M} = J$$

$$[n, n]_n = \mathbb{J}.$$

22/10/2025

Poisson Bracket

u, v functions of (q, p, t) .

$$[u, v]_{p, q} = \frac{\partial u}{\partial q_i} \frac{\partial v}{\partial p_i} - \frac{\partial v}{\partial q_i} \frac{\partial u}{\partial p_i}$$

$$\eta = \begin{pmatrix} q_i \\ p_i \end{pmatrix}$$

$$[u, v]_\eta = \frac{\partial \tilde{u}}{\partial \eta} \mathbb{J} \frac{\partial v}{\partial \eta}.$$

fundamental poisson bracket,

Canonical transformation $\eta \rightarrow \xi$

$$\xi = \xi(\eta) \Rightarrow [\xi, \xi]_\eta = \mathbb{J}.$$

$$\eta = \eta(\xi) \Rightarrow [\eta, \eta]_\xi = \mathbb{J}.$$

Let, two functions u, v of (p, q)

$[u, v]$ is invariant under Canonical Transformation.

$\eta \rightarrow \xi$ (Transformation)

$$[u, v]_\eta = \frac{\partial \tilde{u}}{\partial \eta} \mathbb{J} \frac{\partial v}{\partial \eta}.$$

$$\frac{\partial \tilde{u}}{\partial \eta_j} = \frac{\partial u}{\partial \xi_j} \frac{\partial \xi_j}{\partial \eta_i}$$

$$M_{ij} = \frac{\partial \xi_j}{\partial \eta_i}$$

$$\frac{\partial v}{\partial n_i} = \frac{\partial v}{\partial z_j} \frac{\partial z_j}{\partial n_i}$$

$$= \frac{\partial v}{\partial z_j} M_{ji} = M_{ji} \frac{\partial v}{\partial z_j}$$

$$\widetilde{\frac{\partial v}{\partial n_i}} = \widetilde{M_{ki} \frac{\partial v}{\partial z_k}}$$

$$M_{ji} = \widetilde{M}$$

$$\widetilde{\frac{\partial v}{\partial n}} = \widetilde{M} \frac{\partial v}{\partial z}, \quad \frac{\partial v}{\partial z} M \cdot \widetilde{M} \frac{\partial v}{\partial z} = [v, v]_{\eta}$$

$$[v, v]_{\eta} = \frac{\partial v}{\partial n} \mathbb{J} \frac{\partial v}{\partial n} = \frac{\partial v}{\partial z} \mathbb{J} \frac{\partial v}{\partial z}$$

$$\boxed{M \mathbb{J} \widetilde{M} = \mathbb{J}}$$

$$M = M_{ij}$$

Symplectic geometry

$$\eta \rightarrow \mathbb{J}$$

if η is canonical and if $[\mathbb{J}, \mathbb{J}]_{\eta} = \mathbb{J}$ then $\mathbb{J}$ is canonical.

Properties of Poisson Bracket

① $[v, v] = 0$

② $[v, v] = -[v, v]$

③ $[av + bw, w] = a[v, w] + b[w, w]$

④ $[uv, w] = u[v, w] + [u, w]v$

⑤ $[v, [v, w]] + [v, [w, v]] + [w, [v, v]] = 0$

$$u(q, p, t)$$

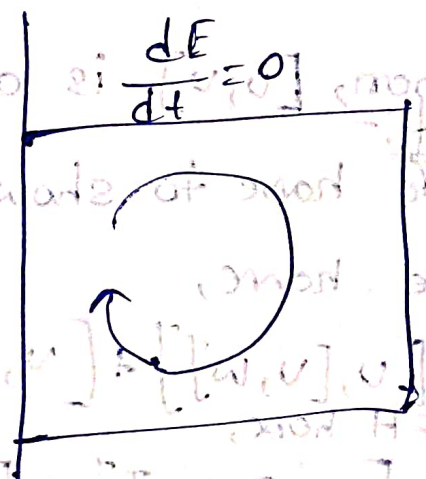
$$\dot{q}_i = \frac{\partial H}{\partial p_i} \quad \text{and} \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

$$\frac{du}{dt} = \frac{\partial u}{\partial q_i} \dot{q}_i + \frac{\partial u}{\partial p_i} \dot{p}_i + \frac{\partial u}{\partial t}$$

$$= \frac{\partial u}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial u}{\partial p_i} \frac{\partial H}{\partial q_i} + \frac{\partial u}{\partial t}$$

$$\frac{du}{dt} = [u, H] + \frac{\partial u}{\partial t}$$

constant of motion means
 E will be constant along that
 trajectory in phase space but
 not constant on neighbouring
 trajectory.



$$u = q_i$$

$$\dot{q}_i = [q_i, H]$$

$$\dot{p}_i = [p_i, H]$$

$$\dot{q}_i = [q_i, H]$$

$$\text{Put } u = H, \quad \frac{dH}{dt} = 0$$

$$\frac{dH}{dt} = [H, H] + \frac{\partial H}{\partial t}$$

$$\Rightarrow \frac{dH}{dt} = \frac{\partial H}{\partial t}$$

Let u be a constant of motion -

$$\frac{du}{dt} = 0 = [u, H] + \frac{\partial u}{\partial t}$$

$$\Rightarrow [u, H] = -\frac{\partial u}{\partial t}$$

if a function is not explicit function of time then it is constant of motion if $[u, H] = 0$.

Let u, v be constants of motion and $\frac{\partial u}{\partial t} = \frac{\partial v}{\partial t} = 0$.

$$\Rightarrow [u, H] = [v, H] = 0$$

then, $[u, v]$ is a constant of motion.

Proof:

We have to show $[H, [u, v]] = 0$

We have,

$$[u, [v, w]] + [v, [w, u]] + [w, [u, v]] = 0$$

$w \equiv H$ here,

$$\Rightarrow [u, [v, H]] + [v, [H, u]] + [H, [u, v]] = 0$$

$$\Rightarrow 0 + 0 + [H, [u, v]] = 0$$

$$\Rightarrow [H, [u, v]] = 0$$

Always we can find 3rd constant of motion if we have two constant of motion. But some time $[u, v] = 0 \rightarrow$ trivial. This is no useful. It is significant when $[u, v]$ is non trivial.

Trivial Example :-

$$H = \frac{p^2}{2m}.$$

We know p_x, p_y are constant of motion.

$$\Rightarrow [p_x, H] = [p_y, H] = 0$$

$$H = \frac{p_x^2 + p_y^2}{2m}.$$

and $[p_x, p_y] = 0 \rightarrow$ from the definition of poisson bracket.

$$H = \frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m} + V(r), \quad r = \sqrt{x^2 + y^2 + z^2}.$$

$$L_x = y p_z - z p_y; \quad L_y = z p_x - x p_z, \quad L_z = x p_y - y p_x.$$

We have to prove $[L_x, H] = 0$.

$$[L_x, H] = [y p_z - z p_y, H] = [y p_z, H] - [z p_y, H].$$

$$\Rightarrow [L_x, H] = [y, H] p_z + y [p_z, H] - z [p_y, H] - [z, H] p_y.$$

$$[y, H] = \dot{y} = \frac{\partial H}{\partial p_y} = \frac{p_y}{m}; \quad \dot{x} = \frac{p_x}{m}, \quad \dot{z} = \frac{p_z}{m}.$$

$$[p_z, H] = \dot{p}_z = \frac{\partial H}{\partial z} = -\frac{\partial V}{\partial z} = -\frac{\partial V}{\partial r} \frac{\partial r}{\partial z} = -\frac{z}{r} \frac{\partial V}{\partial r}.$$

$$[p_x, H] = -\frac{x}{r} \frac{\partial V}{\partial r}, \quad [p_y, H] = -\frac{y}{r} \frac{\partial V}{\partial r}.$$

$$\begin{aligned} \Rightarrow [L_x, H] &= \frac{p_y p_z}{m} - \frac{y z}{r} \frac{\partial V}{\partial r} + \frac{y z}{r} \frac{\partial V}{\partial r} - \frac{p_y p_z}{m} \\ &= 0. \end{aligned}$$

$\Rightarrow L_x$ is constant.

By symmetry of the problem L_x, L_y, L_z are constants.

$$[L_x, L_y] = L_z.$$

Proof:-

$$[L_x, L_y] = [y p_z - z p_y, L_y] \\ = y [p_z, L_y] + [y, L_y] p_z - [z, L_y] p_y - z [p_y, L_y]$$

Homework

Infinitesimal Canonical Transformation

$$Q_i \leftrightarrow q_i$$

$$Q_i = q_i + \delta q_i$$

$$P_i = p_i + \delta p_i$$

$$z_i = \eta_i + \delta \eta_i$$

$$(q, p) \xrightarrow{(\delta q, \delta p)} (Q, P)$$

$\delta q_i, \delta p_i$ are small, ignore $(\delta \eta)^2$.

$$F_2(q, p, t) = G(Q, P, t)$$

$$F_2 = q_i p_i + \epsilon G(q, p, t)$$

$\epsilon \ll 1$ we ignore $\mathcal{O}(\epsilon^2)$.

$$p_i = \frac{\partial F_2}{\partial q_i} \text{ and } Q_i = \frac{\partial F_2}{\partial p_i}$$

$$p_j = p_j^* + \epsilon \frac{\partial G}{\partial q_j}$$

$$\delta p_i = p_j^* - p_j = -\epsilon \frac{\partial G}{\partial q_j}$$

$$\frac{\partial G}{\partial p_j^*} \simeq \frac{\partial G}{\partial p_j}$$

$$Q_j = q_j + \epsilon \frac{\partial G}{\partial p_j}$$

$$\epsilon \frac{\partial G}{\partial p_j^*} = \frac{\partial G}{\partial p_k} \frac{\partial p_k}{\partial p_j^*} = \epsilon \frac{\partial G}{\partial p_k} \begin{pmatrix} 1 & -\epsilon \frac{\partial G}{\partial p_k} \end{pmatrix}$$

$$\simeq \epsilon \frac{\partial G}{\partial p_k}$$

$$\delta \eta = \epsilon \mathbb{J} \frac{\partial G}{\partial \eta}$$

$$\delta p_j^* = -\epsilon \frac{\partial G}{\partial q_j^*}$$

$$\delta q_j^* = \epsilon \frac{\partial G}{\partial p_j^*}$$

$$\begin{array}{c} p \\ \uparrow \\ \text{Q} \end{array}$$

$$(a) \quad \xrightarrow{\quad}$$

$$\varepsilon = dt$$

$$\{t\} \rightarrow \{t_0 + dt\}$$

$$\{t_0\} \rightarrow \{t_0 + dt\}$$

$$M_{ij} = \frac{\partial f_i}{\partial n_j}$$

$$F_2 = q_i p_i + G$$

$$\{n, t_0\} \rightarrow \{n, t\}$$

$$\Rightarrow \{n, t\} = \{n\} + dt(\quad)$$

$$\dot{n} = J \frac{\partial H}{\partial n}$$

$$\frac{n(t) - n(t_0)}{t - t_0} = J \frac{\partial H}{\partial n}$$

$$\Rightarrow n(t) = n(t_0) + dt J \frac{\partial H}{\partial n}$$

$$\{n, t\} = \{n, t_0\} + dt J \frac{\partial H}{\partial n}$$

$$M = \mathbb{1} + \varepsilon J \frac{\partial^2 G}{\partial n^2}$$

check by condition for canonical transformation

$$M J \tilde{M} = J$$

$$\tilde{f}_i = n_i + \varepsilon n_i$$

$$M_{ij} = \frac{\partial f_i}{\partial n_j} \Rightarrow M_{ij} = \delta_{ij} + \frac{\partial}{\partial n_j} (\varepsilon n_i)$$

$$= \delta_{ij} + \varepsilon \frac{\partial}{\partial n_j} J_{ik} \frac{\partial G}{\partial n_k}$$

$$\Rightarrow M_{ij} = \mathbb{1} + \varepsilon J \frac{\partial^2 G}{\partial n_j \partial n_k}$$

$$\dot{n} = [n, H]_n = J \frac{\partial H}{\partial n}$$

$$\left[\because [n, H]_n = J \frac{\partial H}{\partial n} \right]$$

$$\delta n = dt J \frac{\partial H}{\partial n} = dt [n, H]$$

$$\frac{n(t) - n_0}{dt} =$$

$$\dot{n} = [n, H]$$

$$n = n_0 + dt [n, H]$$

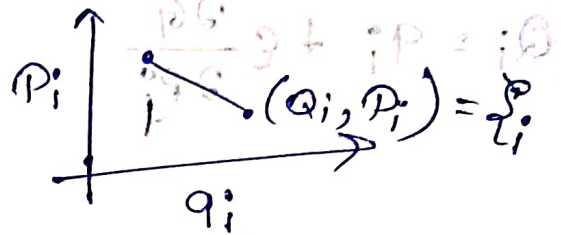
$$\delta n = \int \frac{\partial G}{\partial n} dt = [n, H] dt$$

29/10/2025

Infinitesimal Canonical Transform

$$Q_i = q_i + \delta q_i$$

$$P_i = p_i + \delta p_i$$



$$\delta n < 1 \text{ small } \delta p \text{ and } \delta q \Rightarrow (\delta p)^2 \delta q$$

$$n_i = \begin{pmatrix} q_i \\ p_i \end{pmatrix}$$

$$\dot{x}^i = \dot{x}^i + d\dot{x}^i$$

Symplectic geometry

$$n_i = \begin{pmatrix} q \\ p \end{pmatrix}$$

$$z_i = n_i + \delta n_i$$

$$F_2(q, p, t) = q_i p_i + \epsilon G(q, p, t)$$

$$P_i = \frac{\partial F_2}{\partial q_i} \quad Q_i = \frac{\partial F_2}{\partial p_i}$$

$$P_j = p_j + \epsilon \frac{\partial G}{\partial q_j} \Rightarrow p_j = p_j - \epsilon \frac{\partial G}{\partial q_j}$$

$$\frac{\partial p_j}{\partial p_i} = \delta_{ij} + \epsilon \frac{\partial G}{\partial q_i \partial p_i}$$

$$\delta p_i = -\epsilon \frac{\partial G}{\partial q_i}$$

$$Q_i = q_i + \epsilon \frac{\partial G}{\partial p_i}$$

$$Q_i = q_i + \epsilon \frac{\partial G}{\partial p_j} \frac{\partial p_j}{\partial p_i}$$

$$= q_i + \epsilon \frac{\partial G}{\partial p_j} (\delta_{ij} + \epsilon [J])$$

$$Q_i = q_i + \epsilon \frac{\partial G}{\partial p_i}$$

$$\delta q_i = \epsilon \frac{\partial G}{\partial p_i}$$

$$\delta q_i = \epsilon \frac{\partial G}{\partial p_i}$$

$$\delta p_i = -\epsilon \frac{\partial G}{\partial q_i}$$

$$\tilde{q}_i = q_i + \delta q_i$$

$$\delta m_i = J_{ij} \epsilon \frac{\partial G}{\partial m_j}$$

$$\delta m = J \epsilon \frac{\partial G}{\partial m}$$

Poisson Bracket

$$[u, v]_m = \frac{\partial u}{\partial m} J \frac{\partial v}{\partial m} = J \frac{\partial u}{\partial m} \frac{\partial v}{\partial m}$$

$$u = m$$

$$v = u$$

$$[m, u]_m = J \frac{\partial u}{\partial m}$$

$$[m_i, u] = J_{ij} \frac{\partial u}{\partial m_j}$$

$$\boxed{\delta m_i = \epsilon [m_i, G]_m}$$

Let $G(n, t)$.

$$\tilde{z}_i = z_i + \delta z_i$$

$$\delta z_i = \epsilon [z_i, G]$$

$$\dot{z}_i = J_{ij} \frac{\partial H}{\partial z_j} = [z, H]$$

$$[z, v] = \frac{\partial v}{\partial z} \quad \dot{z}_i = [z_i, H]$$

$$\frac{z_i(t+dt) - z_i(t)}{dt} = [z_i, H]$$

$$z_i(t+dt) = z_i(t) + dt [z_i, H]$$

$$z_i(t) \quad \tilde{z}_i = z_i(t+dt)$$

$$\tilde{z}_i = z_i + dt [z_i, H]$$

$$G(n, t) \quad \delta z_i = \epsilon [z_i, G]$$

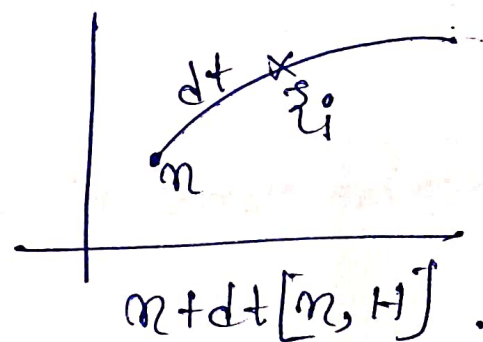
$$\tilde{z}_i = z_i + \delta z_i$$

$$\tilde{z}_i = \tilde{z}_i(z) \rightarrow \text{point transformation}$$

$$M_{ij}^{\alpha\beta} = \frac{\partial \tilde{z}_i^\alpha}{\partial z_j^\beta}$$

$$M J \tilde{M} = J$$

$$\tilde{z}_i = z_i + \epsilon J_{ij} \frac{\partial G}{\partial z_j}$$



$$M_{ij} = \delta_{ij} + \epsilon T_{ik} \frac{\partial^2 G}{\partial m_j \partial m_k}$$

$$[D, \psi] = 0$$

$$[H, \psi] = \frac{116}{100} \psi$$

$$[H, \psi] = \psi$$

$$\frac{100}{100} = [0, \psi]$$

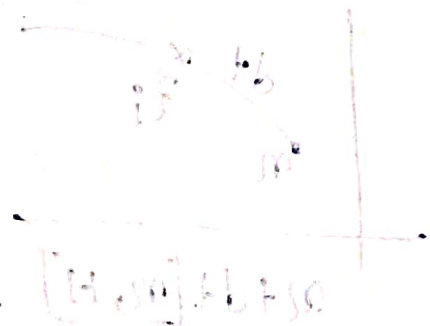
$$[H, \psi] = \frac{(1) \psi - (16+1) \psi}{16}$$

$$[H, \psi] + 16 + (1) \psi = (16+1) \psi$$

$$(16+1) \psi = \psi \quad (1) \psi$$

$$[H, \psi] + 16 + \psi = \psi$$

$$[D, \psi] = 0 \quad (1, \psi)$$



Point + moment of rotation

$$L = \pi U M$$

$$\frac{100}{100} [L + \psi] = \psi$$

Tutorial Problem

1) Consider $\mathcal{L} = \frac{m}{2} \left(\frac{1}{\sigma^2 + \tau^2} \right) [\dot{\sigma}^2 + \dot{\tau}^2]$.

Find $H = ?$

$$\Rightarrow H = \frac{\sigma^2 + \tau^2}{2m} [P_\sigma^2 + P_\tau^2].$$

$$\dot{\sigma} \rightarrow P_\sigma.$$

What is the Hamiltonian EOM :-

$$\dot{P}_\sigma = \frac{\sigma}{m} (P_\sigma^2 + P_\tau^2)$$

$$2) \quad \begin{aligned} q &\rightarrow Q \\ p &\rightarrow P \end{aligned}$$

$$Q = q \cos \alpha - p \sin \alpha$$

$$P = q \sin \alpha + p \cos \alpha,$$

Check wheather Transformation is canonical or not?

$$q = Q \cos \alpha + P \sin \alpha$$

$$p = -Q \sin \alpha + P \cos \alpha.$$

$$M = \begin{pmatrix} \frac{\partial Q}{\partial q} & \frac{\partial Q}{\partial p} \\ \frac{\partial P}{\partial q} & \frac{\partial P}{\partial p} \end{pmatrix} = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$$

$$A^{-1} = \widetilde{A}.$$

$$\det(M) = 1.$$